# EXPERIMENTS

## 1. FERMENTATION BY YEAST

### AIM

To find the variation in the process of fermentation by yeast in the given samples I, II and III.

### MATERIALS REQUIRED

- Glucose solution
- Palm sugar / Jaggery solution
- Jaggery with salt / starch solution
- Yeast solution
- Test tubes
- Delivery tube
- One holed rubber cork
- Lime water

### PRINCIPLE

Fermentation is an anaerobic metabolic process accompanied with effervescence. During this process, sugar is converted into ethyl alcohol and $CO_2$. Yeast has an enzyme zymase which catalyses the fermentation process.

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \xrightarrow[\text{Yeast}]{\text{Zymase}} \underset{\text{Ethyl alcohol}}{2C_2H_5OH} + 2CO_2$$

### PROCEDURE

- Take 2ml of the given samples I, II and III in three clean test tubes (labelled as 1,2 and 3) respectively.
- Add 2ml of yeast solution in all the test tubes and plug the tubes with cotton wool.
- Wait while fermentation takes place and note the time taken.
- Appearance of effervescence in the test tube indicates that fermentation has taken place.
- Remove the cotton wool and pass the gas through a delivery tube into a test tube containing limewater.
- The lime water turns milky indicating that the gas evolved during fermentation is carbon dioxide.

### OBSERVATION

| SL.NO. | SAMPLE | TIME TAKEN | INFERENCE |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

### INFERENCE

The variation in the time taken for fermentation to take place in the different sugar solutions indicates that the simple sugars like glucose are fermented much quicker than the complex sugars.

### PREPARATION OF YEAST SOLUTION

- 100ml of lukewarm distilled water + 10g of yeast granules.



XII Std Zoology Practical EM.indd 252 23/12/2022 16:55:07